

# Dr. Rathin Chandra Shit

Ph.D. in Computer Science & Engineering · IIIT, Bhubaneswar, Odisha, India

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## Research Interests

Edge AI & TinyML · IoT Localization & Indoor Positioning · Federated & Continual Learning · Multi-Robot Task Allocation · Swarm Robotics · Intelligent Transportation Systems · Cyber-Physical Systems Security · Blockchain for Autonomous Systems · Privacy-Preserving Distributed Inference · Wireless Sensor Networks

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## Research Impact

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|----------------------------|--------------------------------------------------------------------------|
| Google Scholar Citations   | 673                                                                      |
| h-index                    | 10                                                                       |
| i10-index                  | 11                                                                       |
| SCI Journal Papers         | 9                                                                        |
| Conference Papers          | 9                                                                        |
| Preprints / Working Papers | 5                                                                        |
| Journals Peer-Reviewed     | 100+ articles across 24 journals                                         |
| TPC Memberships            | 15 (2021–2026)                                                           |
| IEEE CST Impact Factor     | 46.7 (JCR 2024)                                                          |
| Grant Evaluation           | Expert Evaluator, Israel Ministry of Innovation,<br>Science & Technology |

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## Education

**Doctor of Philosophy (Ph.D.) in Computer Science & Engineering** 2016–2021  
*International Institute of Information Technology (IIIT), Bhubaneswar, India* Award: 16 March 2021

- Dissertation: *Localization Algorithms, Strategies and Location-Based Services for Internet of Things*
- Advisor: Prof. Suraj Sharma

**Master of Technology (M.Tech) in Information Technology** 2014–2016  
*Veer Surendra Sai University of Technology (VSSUT), Burla, India* CGPA: 9.08/10.0

- Specialization: Information and Communication Technology
- Thesis: *Effectiveness of Anchor Positioning in Localization of Wireless Sensor Networks*
- Advisor: Prof. Sasmita Behera

**Bachelor of Technology (B.Tech) in Electronics & Telecommunication Engineering** 2006–2010  
*Biju Patnaik University of Technology (BPUT), Rourkela, India* CGPA: 7.67/10.0

- Thesis: *Wireless Missing Person Detection*

## Research & Professional Experience

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### Expert Grant Evaluator (Engineering & Advanced Technology)

2024–Present

Israel Ministry of Innovation, Science and Technology

Remote / National R&D Evaluation

- Evaluate high-stakes AI R&D proposals for national funding consideration.
- Assess technical viability, innovation potential, and funding merit of image-localization and related advanced technology applications.
- Provide expert technical recommendations to guide public investment in emerging AI research directions.

### Founder & Principal Researcher

2021–Present

NavSense Lab (Navigation & Sensing Laboratory), Bhubaneswar, India

- Founded an independent research laboratory focused on edge AI, IoT localization, and autonomous multi-robot coordination.
- Directing a six-paper active submission pipeline: federated learning, AI-blockchain integration, multi-robot task allocation, and urban traffic optimization.
- One Elsevier paper accepted (April 2026, camera-ready); multiple submissions under review at IEEE flagship venues.
- International collaborations: University of Newcastle (UK), University of Sydney (Australia), Central Michigan University (USA), Macquarie University (Australia).
- Open-source presence: GitHub ([navsenselab](#)), Hugging Face ([navsenselab](#)).

### AI/ML R&D Engineer

2020–Present

Coreforce

Global Remote / Industry

- Completed and delivered on **50+ projects** in IoT, Cloud Computing, and AI/ML for international industry clients.

### Ph.D. Researcher

2016–2021

International Institute of Information Technology (IIIT), Bhubaneswar

- Developed IoT localization algorithms addressing GPS outage, NLoS, multi-path, multi-floor, and power-constrained deployments.
- Published 9 SCI-indexed papers; two IEEE CST surveys (IF 46.7) form the core citation base.
- Presented at IEEE ICC 2020 (Dublin, H-index 61) and ICORT 2021.

### Senior Research Fellow

August 2014–August 2016

Defence Research and Development Organisation (DRDO), Govt. of India

Chandipur, India

- Developed PCMC Radar Data Processing Sub-System (DPSS) for real-time target tracking and systematic error correction.
- Designed wideband Ultra-Low Noise Amplifier (LNA); published in *IET Circuits, Devices & Systems* (IF 1.2).
- Designed waveguide components; analyzed C-band radar receiver data.

### Junior Research Fellow

August 2012–August 2014

Defence Research and Development Organisation (DRDO), Govt. of India

Chandipur, India

- Developed Angle Tracking Sub-System (ATSS) software with servo control, error correction, and star calibration.

- Co-authored radar front-end paper in *Defence Science Journal*.

## Teaching Experience

### Graduate Teaching Faculty — IIIT Bhubaneswar

| Period       | Course                                    | Level         |
|--------------|-------------------------------------------|---------------|
| Jan–Jul 2019 | Data Communication & Computer Network Lab | Undergraduate |
| Aug–Dec 2018 | Computer Organization Lab                 | Undergraduate |
| Aug–Dec 2017 | Object Oriented Programming Lab (C++)     | Undergraduate |
| Aug–Dec 2016 | Artificial Intelligence                   | Undergraduate |

### Guest Lectures & Outreach

- Jan 2020 — Cyber Safety & Security Awareness Program, IIIT Bhubaneswar
- Feb–Mar 2019 — Introduction to Python Programming, Navodaya Leadership Institute, Konark (Ministry of HRD, Govt. of India)
- Jan–Feb 2017 — Computer Courses, Biju Patnaik State Police Academy, Bhubaneswar

### Invited FDP/TTP Speaker

- **AICTE-CSVTU MoU Joint Teachers Training Program (TTP-2021)** on *SMART PERSPECTIVES OF IoT IN DIFFERENT APPLICATIONS*, hosted by Shri Shankaracharya Technical Campus, Bhilai, Chhattisgarh (10–15 December 2021).
- Role: **Invited Speaker**; *Presented* Topic 1: *IoV for Connected Vehicles* Topic 2: *Localization Techniques in IoT*.

## Publications

Google Scholar: 673 citations · h-index: 10 · i10-index: 11 · as of May 2026

### Journal Articles (SCI-Indexed, ranked by Impact Factor)

1. R. C. Shit, S. Sharma, D. Puthal, P. James, B. Pradhan, A. van Moorsel, A. Y. Zomaya, and R. Ranjan, “Ubiquitous Localization (UbiLoc): A Survey and Taxonomy on Device Free Localization for Smart World,” *IEEE Communications Surveys & Tutorials*, vol. 21, no. 4, pp. 3532–3564, 2019. [IF: 46.7] [SCI]  
doi:10.1109/COMST.2019.2915923
2. R. C. Shit, S. Sharma, D. Puthal, and A. Y. Zomaya, “Location of Things (LoT): A Review and Taxonomy of Sensors Localization in IoT Infrastructure,” *IEEE Communications Surveys & Tutorials*, vol. 20, no. 3, pp. 2028–2061, 2018. [IF: 46.7] [SCI]  
doi:10.1109/COMST.2018.2798591
3. R. C. Shit, S. Sharma, K. Yelamarthi, and D. Puthal, “AI-Enabled Fingerprinting and Crowdsourced Vehicle Localization for Resilient and Safe Transportation Systems,” *IEEE Transactions on Intelligent Transportation Systems*, vol. 22, no. 7, pp. 4660–4669, Jul. 2021. [IF: 8.4] [SCI]  
doi:10.1109/TITS.2021.3053942
4. R. C. Shit, “Precise Localization for Achieving Next-Generation Autonomous Navigation: State-of-the-Art, Taxonomy and Future Prospects,” *Computer Communications*, vol. 160, pp. 351–374, Jul. 2020. [IF: 4.3] [SCI]  
doi:10.1016/j.comcom.2020.06.007
5. R. C. Shit, “Crowd Intelligence for Sustainable Futuristic Intelligent Transportation System: A

- Review,” *IET Intelligent Transport Systems*, vol. 14, no. 6, pp. 480–494, Apr. 2020. [IF: 2.9] [SCI]  
doi:10.1049/iet-its.2019.0321
6. R. C. Shit and S. Sharma, “Ray-Tracing Assisted Fingerprinting for Localization in IoT Health 4.0,” *Transactions on Emerging Telecommunications Technologies*, vol. 33, no. 11, Jun. 2022. [IF: 2.5] [SCI]  
doi:10.1002/ett.4573
  7. R. C. Shit, S. Sharma, P. Watters, K. Yelamarthi, B. Pradhan, R. Davison, G. Morgan, and D. Puthal, “Privacy-Preserving Cooperative Localization in Vehicular Edge Computing Infrastructure,” *Concurrency and Computation: Practice and Experience*, vol. 34, no. 14, Jun. 2020. [IF: 1.5] [SCI]  
doi:10.1002/cpe.5827
  8. A. K. Ray and R. C. Shit, “Design of Ultra-Low Noise, Wideband Low-Noise Amplifier for Highly Survival Radar Receiver,” *IET Circuits, Devices & Systems*, vol. 10, no. 6, pp. 473–480, Nov. 2016. [IF: 1.2] [SCI]  
doi:10.1049/iet-cds.2016.0065
  9. A. K. Ray, R. C. Shit, and R. S. Yadav, “Development of C-Band RF Front-End of Precision Coherent Mono-Pulse C-Band Radar,” *Defence Science Journal*, vol. 64, no. 4, pp. 358–365, Jul. 2014. [IF: 0.7] [SCI]  
doi:10.14429/dsj.64.4245

#### Conference Proceedings (Peer-Reviewed, ranked by venue prestige)

1. R. C. Shit, S. Sharma, M. S. Obaidat, and D. Puthal, “Adaptive Software Defined Node Deployment for Green Internet of Things,” in *ICC 2020 — IEEE International Conference on Communications*, Dublin, Ireland, Jun. 2020. [IEEE ICC · H-index 61 · Flagship IEEE ComSoc]  
doi:10.1109/ICC40277.2020.9149396
2. R. C. Shit and S. Subudhi, “Hierarchical Federated Graph Attention Networks for Scalable and Resilient UAV Collision Avoidance,” in *2025 IEEE International Conference on Advanced Networks and Telecommunications Systems (ANTS)*, Dec. 2025. [IEEE Xplore: 16 Mar 2026]  
doi:10.1109/ANTS66931.2025.11430069
3. R. C. Shit and S. Subudhi, “AI-Powered Anomaly Detection with Blockchain for Real-Time Security and Reliability in Autonomous Vehicles,” in *2025 IEEE Space, Aerospace and Defence Conference (SPACE)*, Jul. 2025.
4. R. C. Shit and M. Goswami, “Blockchain Use Case in Multi-Sensor Satellite Architecture,” in *2021 2nd IEEE International Conference on Range Technology (ICORT)*, Aug. 2021.  
doi:10.1109/ICORT52730.2021.9581969
5. R. C. Shit, S. Sharma, K. N. Singh, and S. S. Tripathi, “Overview and Perspective of Localization Accuracy for Persistent Autonomous Vehicle Systems,” in *2019 IEEE ICIT*, Dec. 2019, pp. 211–216.  
doi:10.1109/ICIT48102.2019.00044
6. M. R. Dey, S. Sharma, R. C. Shit, C. P. Meher, and H. K. Pati, “IoV-Based Real-Time Smart Traffic Monitoring System for Smart Cities Using Augmented Reality,” in *2019 IEEE ViTECoN*, Mar. 2019.  
doi:10.1109/ViTECoN.2019.8899362
7. R. C. Shit, S. Sharma, D. Puthal, and S. S. Tripathi, “Probabilistic RSS Fingerprinting for Localization in Smart Platforms,” in *2018 IEEE ICIT*, Dec. 2018, pp. 254–259.  
doi:10.1109/ICIT.2018.00059
8. R. C. Shit and S. Sharma, “Localization for Autonomous Vehicle,” in *2018 IEEE AESPC*, Oct. 2018.  
doi:10.1109/AESPC44649.2018.9033329

9. R. C. Shit, S. Sharma, D. Puthal, and A. Panday, "Self Deployment Based on Circle Packing Algorithm for Movement Assisted Wireless Sensor Networks," in *2017 IEEE ICIT*, Dec. 2017, pp. 240–245.  
doi:10.1109/ICIT.2017.11

## Preprints & Working Papers

- [P1] R. C. Shit and S. Subudhi, "Hierarchical Adaptive Consensus Network: A Dynamic Framework for Scalable Consensus in Collaborative Multi-Agent AI Systems," 2025. (*Under review*)
- [P2] R. C. Shit and S. Subudhi, "Scalable Hierarchical AI-Blockchain Framework for Real-Time Anomaly Detection in Large-Scale Autonomous Vehicle Networks," 2025. (*Under review*)
- [P3] R. C. Shit and S. Subudhi, "Privacy-Preserving Federated Learning for Fair and Efficient Urban Traffic Optimization," 2025. (*Under review*)
- [P4] R. C. Shit, "Path-Coordinated Continual Learning with Neural Tangent Kernel-Justified Plasticity," arXiv, 2025.
- [P5] R. C. Shit and S. Subudhi, "Multi-Robot Task Allocation for Homogeneous Tasks with Collision Avoidance via Spatial Clustering," arXiv:2505.10073, 2025.  
doi:10.48550/arXiv.2505.10073

## Honors & Awards

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| Year      | Award                      | Awarding Body        |
|-----------|----------------------------|----------------------|
| 2014–2016 | Senior Research Fellowship | DRDO, Govt. of India |
| 2012–2014 | Junior Research Fellowship | DRDO, Govt. of India |
| 2013      | GATE Qualified             | MHRD, Govt. of India |
| 2010      | GATE Qualified             | MHRD, Govt. of India |

## Academic Service

### Conference Technical Program Committee (TPC)

2026 IEEE SPACE 2026; IEEE ANTS 2026; ISGAIE 2026  
 2025 IEEE ANTS 2025; IEEE ANTS 2025 WiE Track; IEEE SPACE 2025;  
 ICORT 2025; ISGAIE 2025  
 2023 ICORT 2023  
 2022 GAICES 2022; ICATH 2022; ICCCSET 2022; IFC 2022  
 2021 ICORT 2021

**Book Editorial:** TPC Member, *Combating Fake News with Computational Intelligence Techniques*, Springer Studies in Computational Intelligence (Scopus), 2021.

### Peer Review — 100+ Articles, 24 Journals

#### IEEE:

- IEEE Trans. on Intelligent Transportation Systems
- IEEE Open Journal of the Computer Society (*2026*)
- IEEE Trans. on Mobile Computing
- IEEE Sensors Journal
- IEEE Industrial Electronics Magazine
- IEEE Trans. on Automation Science & Engg.

## Elsevier / Springer / Nature:

- Engineering Applications of AI — 8 reviews
- Computer Communications — 6 reviews
- Ad Hoc Networks — 6 reviews
- Intl. J. Control, Automation & Systems — 7 reviews
- Journal of Network & Computer Applications —

5 reviews

- Expert Systems with Applications — 4 reviews
- Intl. J. Advanced Robotic Systems — 6 reviews
- Intl. J. Computational Intelligence Systems — 3 reviews
- Scientific Reports (Nature Portfolio) — 1 review
- JMIR — 2 reviews

Books Reviewed: 1 (Springer)

## Technical Skills

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|----------------------|-------------------------------------------------------------------------------------------------------|
| <b>Programming</b>   | Python, C, C++, MATLAB                                                                                |
| <b>AI / ML</b>       | TensorFlow, PyTorch, scikit-learn; CNNs, RNNs, Graph Attention Networks, NTK-based Continual Learning |
| <b>Simulation</b>    | NetworkX, DeepGraph, MATLAB/Simulink                                                                  |
| <b>RF / Hardware</b> | ADS, CST Microwave Studio, HFSS, System Vue, LabVIEW                                                  |
| <b>Technologies</b>  | IoT, Edge Computing, Federated Learning, Blockchain, WSN, UAV                                         |
| <b>Open Source</b>   | GitHub: <a href="#">navsenselab</a> · Hugging Face: <a href="#">navsenselab</a>                       |
| <b>Documentation</b> | L <sup>A</sup> T <sub>E</sub> X, Microsoft Office Suite                                               |

## Research Vision

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The research question that I am working on is related to the development of an edge-deployed AI system capable of sensing, localization, coordinating and reasoning reliably, privately and autonomously without depending on the cloud. My research program is guided by three converging approaches:

1. **Edge-Native Localization for Pervasive IoT.** Lightweight, privacy-aware positioning in GPS-denied and NLoS-challenged environments using AI fingerprinting, ray-tracing and crowdsourced RSSI. Based on two groundbreaking IEEE CST surveys (673 citations) that establish a taxonomy of the field.
2. **Continual & Federated Learning for Safety-Critical Autonomy.** Neural Tangent Kernel-justified frameworks for plasticity-stability balance in continual learning, extended to federated graph attention networks for scalable, privacy-preserving coordination across UAV, vehicle, and multi-robot networks.
3. **Secure Multi-Agent Autonomy via AI-Blockchain Integration.** Hierarchical AI-blockchain architectures and anomaly-detection pipelines to strengthen real-time decision-making in large-scale multi-sensor satellite systems and autonomous vehicle networks against adversarial attacks and Byzantine faults.

## References

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*References available upon request.*

Last Updated: May 2026